

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech

Date: 11-03-2026

Faculty Name: Prof. S.Gopikrishnan

School: SCOPE

Student name: J MADHAN KUMAAR

Reg. no.: 23BCE8930

1. Use MongoDB to implement the following DB operations

1. Create a database called 'vehicles' and *write* a MongoDB query to select database as "vehicles".

```
Welcome  mongosh: localhost:27017  vehicles  localhost:27017  +
>_MONGOSH
> use vehicles
< switched to db vehicles
```

2. Write a MongoDB query to display all the databases.

```
> show dbs
< admin    40.00 KiB
  config   12.00 KiB
  local    64.00 KiB
```

3. Create a collection called 'two_wheelers'. (use capping) and Create a collection called 'four_wheelers'.

```
> db.createCollection("two_wheelers", { capped: true, size: 10000000, max: 1000 })
< { ok: 1 }
> db.createCollection("four_wheelers")
< { ok: 1 }
```

4. Add 5 two-wheeler details to the collection named 'two_wheelers'. Each document consists of following fields as bike_name, model (gear or gearless), category (100cc, 125cc, 150cc, 200cc), colors_available (red, black, blue, sport red etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.

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```
> db.two_wheelers.insertMany([
  {
    bike_name: "Honda Shine",
    model: "gear",
    category: "125cc",
    colors_available: ["red", "black", "blue"],
    manufacturer: "Honda",
    performance: 7,
    timestamp: new Date("2022-01-15"),
    price: 75000
  },
  {
    bike_name: "TVS Jupiter",
    model: "gearless",
    category: "110cc",
    colors_available: ["white", "grey", "blue"],
    manufacturer: "TVS",
    performance: 8,
    timestamp: new Date("2021-06-10"),
    price: 72000
  },
```

```
{
  bike_name: "Yamaha R15",
  model: "gear",
  category: "150cc",
  colors_available: ["sport red", "black", "blue"],
  manufacturer: "Yamaha",
  performance: 9,
  timestamp: new Date("2023-03-20"),
  price: 165000
},
{
  bike_name: "Royal Enfield Classic 350",
  model: "gear",
  category: "350cc",
  colors_available: ["black", "green", "brown"],
  manufacturer: "Royal Enfield",
  performance: 8,
  timestamp: new Date("2022-11-05"),
  price: 192000
},
```

```
{
  bike_name: "Activa 6G",
  model: "gearless",
  category: "125cc",
  colors_available: ["red", "black", "pearl white"],
  manufacturer: "Honda",
  performance: 7,
  timestamp: new Date("2020-08-01"),
  price: 74000
}
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69b13809a03ea65b4b9716f6'),
    '1': ObjectId('69b13809a03ea65b4b9716f7'),
    '2': ObjectId('69b13809a03ea65b4b9716f8'),
    '3': ObjectId('69b13809a03ea65b4b9716f9'),
    '4': ObjectId('69b13809a03ea65b4b9716fa')
  }
}
```

5. Add 5 four-wheeler details to the collection named 'four_wheelers'. Each document consists of following fields as vehicle_name, model (commercial or own), category (car, lorry, bus, mini truck, heavy truck, containers), variants (vxi, zxi, petrol, diesel etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.

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```
>_MONGOSH
> db.four_wheelers.insertMany([
  {
    vehicle_name: "Maruti Swift",
    model: "own",
    category: "car",
    variants: ["vxi", "zxi", "petrol", "diesel"],
    manufacturer: "Maruti Suzuki",
    performance: 8,
    timestamp: new Date("2022-04-10"),
    price: 650000
  },
  {
    vehicle_name: "Tata Nexon",
    model: "own",
    category: "car",
    variants: ["petrol", "diesel", "electric"],
    manufacturer: "Tata Motors",
    performance: 9,
    timestamp: new Date("2023-01-15"),
    price: 900000
  },
```

```
{
  vehicle_name: "Ashok Leyland Dost",
  model: "commercial",
  category: "mini truck",
  variants: ["diesel"],
  manufacturer: "Ashok Leyland",
  performance: 7,
  timestamp: new Date("2021-07-20"),
  price: 750000
},
{
  vehicle_name: "Volvo B11R",
  model: "commercial",
  category: "bus",
  variants: ["diesel"],
  manufacturer: "Volvo",
  performance: 9,
  timestamp: new Date("2020-11-30"),
  price: 5000000
},
```

```
{
  vehicle_name: "Mahindra Bolero",
  model: "own",
  category: "car",
  variants: ["diesel", "zxi"],
  manufacturer: "Mahindra",
  performance: 6,
  timestamp: new Date("2021-09-05"),
  price: 980000
}
})
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69b13814a03ea65b4b9716fb'),
    '1': ObjectId('69b13814a03ea65b4b9716fc'),
    '2': ObjectId('69b13814a03ea65b4b9716fd'),
    '3': ObjectId('69b13814a03ea65b4b9716fe'),
    '4': ObjectId('69b13814a03ea65b4b9716ff')
  }
}
```

6. Write a MongoDB query to display all documents available in two_wheelers and four_wheelers.

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```
> db.two_wheelers.find().pretty()
< {
  _id: ObjectId('69b13809a03ea65b4b9716f6'),
  bike_name: 'Honda Shine',
  model: 'gear',
  category: '125cc',
  colors_available: [
    'red',
    'black',
    'blue'
  ],
  manufacturer: 'Honda',
  performance: 7,
  timestamp: 2022-01-15T00:00:00.000Z,
  price: 75000
}
{
  _id: ObjectId('69b13809a03ea65b4b9716f7'),
  bike_name: 'TVS Jupiter',
  model: 'gearless',
  category: '110cc',
  colors_available: [
    'white',
    'grey',
    'blue'
  ],
  manufacturer: 'TVS',
  performance: 8,
  timestamp: 2021-06-10T00:00:00.000Z,
  price: 72000
}
{
  _id: ObjectId('69b13809a03ea65b4b9716f8'),
  bike_name: 'Yamaha R15',
  model: 'gear',
  category: '150cc',
  colors_available: [
    'sport red',
    'black',
    'blue'
  ],
  manufacturer: 'Yamaha',
  performance: 9,
  timestamp: 2023-03-20T00:00:00.000Z,
  price: 165000
}
{
  _id: ObjectId('69b13809a03ea65b4b9716f9'),
  bike_name: 'Royal Enfield Classic 350',
  model: 'gear',
  category: '350cc',
  colors_available: [
    'black',
    'green',
    'brown'
  ],
  manufacturer: 'Royal Enfield',
  performance: 8,
  timestamp: 2022-11-05T00:00:00.000Z,
  price: 192000
}
{
  _id: ObjectId('69b13809a03ea65b4b9716fa'),
  bike_name: 'Activa 6G',
  model: 'gearless',
  category: '125cc',
  colors_available: [
    'red',
    'black',
    'pearl white'
  ],
  manufacturer: 'Honda',
  performance: 7,
  timestamp: 2020-08-01T00:00:00.000Z,
  price: 74000
}
```

```
> db.two_wheelers.find().pretty()
< {
  _id: ObjectId('69b13809a03ea65b4b9716f6'),
  bike_name: 'Honda Shine',
  model: 'gear',
  category: '125cc',
  colors_available: [
    'red',
    'black',
    'blue'
  ],
  manufacturer: 'Honda',
  performance: 7,
  timestamp: 2022-01-15T00:00:00.000Z,
  price: 75000
}
{
  _id: ObjectId('69b13809a03ea65b4b9716f7'),
  bike_name: 'TVS Jupiter',
  model: 'gearless',
  category: '110cc',
  colors_available: [
    'white',
    'grey',
    'blue'
  ],
  manufacturer: 'TVS',
  performance: 8,
  timestamp: 2021-06-10T00:00:00.000Z,
  price: 72000
}
{
  _id: ObjectId('69b13809a03ea65b4b9716f8'),
  bike_name: 'Yamaha R15',
  model: 'gear',
  category: '150cc',
  colors_available: [
    'sport red',
    'black',
    'blue'
  ],
  manufacturer: 'Yamaha',
  performance: 9,
  timestamp: 2023-03-20T00:00:00.000Z,
  price: 165000
}
{
  _id: ObjectId('69b13809a03ea65b4b9716f9'),
  bike_name: 'Royal Enfield Classic 350',
  model: 'gear',
  category: '350cc',
  colors_available: [
    'black',
    'green',
    'brown'
  ],
  manufacturer: 'Royal Enfield',
  performance: 8,
  timestamp: 2022-11-05T00:00:00.000Z,
  price: 192000
}
{
  _id: ObjectId('69b13809a03ea65b4b9716fa'),
  bike_name: 'Activa 6G',
  model: 'gearless',
  category: '125cc',
  colors_available: [
    'red',
    'black',
    'pearl white'
  ],
  manufacturer: 'Honda',
  performance: 7,
  timestamp: 2020-08-01T00:00:00.000Z,
  price: 74000
}
```

```
> db.two_wheelers.find().pretty()
< {
  _id: ObjectId('69b13809a03ea65b4b9716f6'),
  bike_name: 'Honda Shine',
  model: 'gear',
  category: '125cc',
  colors_available: [
    'red',
    'black',
    'blue'
  ],
  manufacturer: 'Honda',
  performance: 7,
  timestamp: 2022-01-15T00:00:00.000Z,
  price: 75000
}
{
  _id: ObjectId('69b13809a03ea65b4b9716f7'),
  bike_name: 'TVS Jupiter',
  model: 'gearless',
  category: '110cc',
  colors_available: [
    'white',
    'grey',
    'blue'
  ],
  manufacturer: 'TVS',
  performance: 8,
  timestamp: 2021-06-10T00:00:00.000Z,
  price: 72000
}
{
  _id: ObjectId('69b13809a03ea65b4b9716f8'),
  bike_name: 'Yamaha R15',
  model: 'gear',
  category: '150cc',
  colors_available: [
    'sport red',
    'black',
    'blue'
  ],
  manufacturer: 'Yamaha',
  performance: 9,
  timestamp: 2023-03-20T00:00:00.000Z,
  price: 165000
}
{
  _id: ObjectId('69b13809a03ea65b4b9716f9'),
  bike_name: 'Royal Enfield Classic 350',
  model: 'gear',
  category: '350cc',
  colors_available: [
    'black',
    'green',
    'brown'
  ],
  manufacturer: 'Royal Enfield',
  performance: 8,
  timestamp: 2022-11-05T00:00:00.000Z,
  price: 192000
}
{
  _id: ObjectId('69b13809a03ea65b4b9716fa'),
  bike_name: 'Activa 6G',
  model: 'gearless',
  category: '125cc',
  colors_available: [
    'red',
    'black',
    'pearl white'
  ],
  manufacturer: 'Honda',
  performance: 7,
  timestamp: 2020-08-01T00:00:00.000Z,
  price: 74000
}
```

```
> db.two_wheelers.find().pretty()
< {
  _id: ObjectId('69b13809a03ea65b4b9716f6'),
  bike_name: 'Honda Shine',
  model: 'gear',
  category: '125cc',
  colors_available: [
    'red',
    'black',
    'blue'
  ],
  manufacturer: 'Honda',
  performance: 7,
  timestamp: 2022-01-15T00:00:00.000Z,
  price: 75000
}
{
  _id: ObjectId('69b13809a03ea65b4b9716f7'),
  bike_name: 'TVS Jupiter',
  model: 'gearless',
  category: '110cc',
  colors_available: [
    'white',
    'grey',
    'blue'
  ],
  manufacturer: 'TVS',
  performance: 8,
  timestamp: 2021-06-10T00:00:00.000Z,
  price: 72000
}
{
  _id: ObjectId('69b13809a03ea65b4b9716f8'),
  bike_name: 'Yamaha R15',
  model: 'gear',
  category: '150cc',
  colors_available: [
    'sport red',
    'black',
    'blue'
  ],
  manufacturer: 'Yamaha',
  performance: 9,
  timestamp: 2023-03-20T00:00:00.000Z,
  price: 165000
}
{
  _id: ObjectId('69b13809a03ea65b4b9716f9'),
  bike_name: 'Royal Enfield Classic 350',
  model: 'gear',
  category: '350cc',
  colors_available: [
    'black',
    'green',
    'brown'
  ],
  manufacturer: 'Royal Enfield',
  performance: 8,
  timestamp: 2022-11-05T00:00:00.000Z,
  price: 192000
}
{
  _id: ObjectId('69b13809a03ea65b4b9716fa'),
  bike_name: 'Activa 6G',
  model: 'gearless',
  category: '125cc',
  colors_available: [
    'red',
    'black',
    'pearl white'
  ],
  manufacturer: 'Honda',
  performance: 7,
  timestamp: 2020-08-01T00:00:00.000Z,
  price: 74000
}
```

```
> db.four_wheelers.find().pretty()
< {
  _id: ObjectId('69b13814a03ea65b4b9716fb'),
  vehicle_name: 'Maruti Swift',
  model: 'own',
  category: 'car',
  variants: [
    'vxi',
    'zxi',
    'petrol',
    'diesel'
  ],
  manufacturer: 'Maruti Suzuki',
  performance: 8,
  timestamp: 2022-04-10T00:00:00.000Z,
  price: 650000
}
{
  _id: ObjectId('69b13814a03ea65b4b9716fc'),
  vehicle_name: 'Tata Nexon',
  model: 'own',
  category: 'car',
  variants: [
    'petrol',
    'diesel',
    'electric'
  ],
  manufacturer: 'Tata Motors',
  performance: 9,
  timestamp: 2022-04-10T00:00:00.000Z,
  price: 650000
}
```

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```
{
  manufacturer: 'Tata Motors',
  performance: 9,
  timestamp: 2023-01-15T00:00:00.000Z,
  price: 900000
}
{
  _id: ObjectId('69b13814a03ea65b4b9716fd'),
  vehicle_name: 'Ashok Leyland Dost',
  model: 'commercial',
  category: 'mini truck',
  variants: [
    'diesel'
  ],
  manufacturer: 'Ashok Leyland',
  performance: 7,
  timestamp: 2021-07-20T00:00:00.000Z,
  price: 750000
}
{
  _id: ObjectId('69b13814a03ea65b4b9716fe'),
  vehicle_name: 'Volvo B11R',
  model: 'commercial',
  category: 'bus',
  variants: [
    'diesel'
  ],
  manufacturer: 'Volvo',
  performance: 9,
```

```
manufacturer: 'Volvo',
performance: 9,
timestamp: 2020-11-30T00:00:00.000Z,
price: 5000000
}
{
  _id: ObjectId('69b13814a03ea65b4b9716ff'),
  vehicle_name: 'Mahindra Bolero',
  model: 'own',
  category: 'car',
  variants: [
    'diesel',
    'zxi'
  ],
  manufacturer: 'Mahindra',
  performance: 6,
  timestamp: 2021-09-05T00:00:00.000Z,
  price: 980000
}
```

7. Write a MongoDB query to display only vehicle name and price in all the collection of the database

```
> db.two_whealers.find({}, { bike_name: 1, price: 1, _id: 0 })
< {
  bike_name: 'Honda Shine',
  price: 75000
}
{
  bike_name: 'TVS Jupiter',
  price: 72000
}
{
  bike_name: 'Yamaha R15',
  price: 165000
}
{
  bike_name: 'Royal Enfield Classic 350',
  price: 192000
}
{
  bike_name: 'Activa 6G',
  price: 74000
}
```

```
> db.four_whealers.find({}, { vehicle_name: 1, price: 1, _id: 0 })
< {
  vehicle_name: 'Maruti Swift',
  price: 650000
}
{
  vehicle_name: 'Tata Nexon',
  price: 900000
}
{
  vehicle_name: 'Ashok Leyland Dost',
  price: 750000
}
{
  vehicle_name: 'Volvo B11R',
  price: 5000000
}
{
  vehicle_name: 'Mahindra Bolero',
  price: 980000
}
```

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8. Write a MongoDB query to display two_wheelers from a particular company

```
> db.two_wheelers.find({ manufacturer: "Honda" })
< {
  _id: ObjectId('69b13809a03ea65b4b9716f6'),
  bike_name: 'Honda Shine',
  model: 'gear',
  category: '125cc',
  colors_available: [
    'red',
    'black',
    'blue'
  ],
  manufacturer: 'Honda',
  performance: 7,
  timestamp: 2022-01-15T00:00:00.000Z,
  price: 75000
}
{
  _id: ObjectId('69b13809a03ea65b4b9716fa'),
  bike_name: 'Activa 6G',
  model: 'gearless',
  category: '125cc',
  colors_available: [
    'red',
    'black',
    'pearl white'
  ],
  manufacturer: 'Honda',
  performance: 7,
  timestamp: 2020-08-01T00:00:00.000Z,
```

9. Write a MongoDB query to display four_wheelers available in diesel variants

```
> db.four_wheelers.find({ variants: "diesel" })
< {
  _id: ObjectId('69b13814a03ea65b4b9716fb'),
  vehicle_name: 'Maruti Swift',
  model: 'own',
  category: 'car',
  variants: [
    'vxi',
    'zxi',
    'petrol',
    'diesel'
  ],
  manufacturer: 'Maruti Suzuki',
  performance: 8,
  timestamp: 2022-04-10T00:00:00.000Z,
  price: 650000
}
{
  _id: ObjectId('69b13814a03ea65b4b9716fc'),
  vehicle_name: 'Tata Nexon',
  model: 'own',
  category: 'car',
  variants: [
    'petrol',
    'diesel',
    'electric'
  ],
  manufacturer: 'Tata Motors',
  performance: 9,
```

```
performance: 9,
timestamp: 2023-01-15T00:00:00.000Z,
price: 900000
}
{
  _id: ObjectId('69b13814a03ea65b4b9716fd'),
  vehicle_name: 'Ashok Leyland Dost',
  model: 'commercial',
  category: 'mini truck',
  variants: [
    'diesel'
  ],
  manufacturer: 'Ashok Leyland',
  performance: 7,
  timestamp: 2021-07-20T00:00:00.000Z,
  price: 750000
}
{
  _id: ObjectId('69b13814a03ea65b4b9716fe'),
  vehicle_name: 'Volvo B11R',
  model: 'commercial',
  category: 'bus',
  variants: [
    'diesel'
  ],
  manufacturer: 'Tata Motors',
  performance: 9,
```

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```
{
  _id: ObjectId('69b13814a03ea65b4b9716fe'),
  vehicle_name: 'Volvo B11R',
  model: 'commercial',
  category: 'bus',
  variants: [
    'diesel'
  ],
  manufacturer: 'Volvo',
  performance: 9,
  timestamp: 2020-11-30T00:00:00.000Z,
  price: 5000000
}
{
  _id: ObjectId('69b13814a03ea65b4b9716ff'),
  vehicle_name: 'Mahindra Bolero',
  model: 'own',
  category: 'car',
  variants: [
    'diesel',
    'zxi'
  ],
  manufacturer: 'Mahindra',
  performance: 6,
  timestamp: 2021-09-05T00:00:00.000Z,
  price: 980000
}
```

10. Write a MongoDB query to display vehicles name, category and manufacturer details whose rating is more than 5.

```
>_MONGOSH
> db.two_wheelers.find(
  { performance: { $gt: 5 } },
  { bike_name: 1, category: 1, manufacturer: 1, _id: 0 }
)
< {
  bike_name: 'Honda Shine',
  category: '125cc',
  manufacturer: 'Honda'
}
{
  bike_name: 'TVS Jupiter',
  category: '110cc',
  manufacturer: 'TVS'
}
{
  bike_name: 'Yamaha R15',
  category: '150cc',
  manufacturer: 'Yamaha'
}
{
  bike_name: 'Royal Enfield Classic 350',
  category: '350cc',
  manufacturer: 'Royal Enfield'
}
{
  bike_name: 'Activa 6G',
  category: '125cc',
  manufacturer: 'Honda'
}
```

```
> db.four_wheelers.find(
  { performance: { $gt: 5 } },
  { vehicle_name: 1, category: 1, manufacturer: 1, _id: 0 }
)
< {
  vehicle_name: 'Maruti Swift',
  category: 'car',
  manufacturer: 'Maruti Suzuki'
}
{
  vehicle_name: 'Tata Nexon',
  category: 'car',
  manufacturer: 'Tata Motors'
}
{
  vehicle_name: 'Ashok Leyland Dost',
  category: 'mini truck',
  manufacturer: 'Ashok Leyland'
}
{
  vehicle_name: 'Volvo B11R',
  category: 'bus',
  manufacturer: 'Volvo'
}
{
  vehicle_name: 'Mahindra Bolero',
  category: 'car',
  manufacturer: 'Mahindra'
}
```

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2. Use MongoDB to implement the following DB operations for a Zoo

1. Create a database called 'animal' and *write* a MongoDB query to select database as 'animal'.

```
> use animal
< switched to db animal
```

2. Write a MongoDB query to display all the databases.

```
> show dbs
< admin      40.00 KiB
  config     108.00 KiB
  local      64.00 KiB
  vehicles   80.00 KiB
animal> |
```

3. Create a collection called 'wild_animals' (use capping) and Create a collection called 'domestic_animals'.

```
> db.createCollection("wild_animals", { capped: true, size: 10000000, max: 1000 })
< { ok: 1 }
> db.createCollection("domestic_animals")
< { ok: 1 }
animal> |
```

4. Add 5 wild_animal details to the collection named 'wild_animals'. Each document consists of following fields as animal_name, nature (harm or harmless), favorite_foods (meat, rabbits, deer etc) as array, care_taker_name, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech

Faculty Name: Prof. S.Gopikrishnan

Student name: J MADHAN KUMAAR

Date: 11-03-2026

School: SCOPE

Reg. no.: 23BCE8930

```
>_MONGOSH
> db.wild_animals.insertMany([
  {
    animal_name: "Lion",
    nature: "harm",
    favorite_foods: ["meat", "deer", "zebra"],
    care_taker_name: "John",
    life_span: 15,
    timestamp: new Date("2021-03-10"),
    expenses: 50000
  },
  {
    animal_name: "Tiger",
    nature: "harm",
    favorite_foods: ["meat", "rabbits", "deer"],
    care_taker_name: "Ravi",
    life_span: 18,
    timestamp: new Date("2020-07-22"),
    expenses: 60000
  },
  {
    animal_name: "Elephant",
    nature: "harmless",
    favorite_foods: ["sugarcane", "bananas", "grass"],
    care_taker_name: "John",
    life_span: 60,
    timestamp: new Date("2019-11-05"),
    expenses: 80000
  },
],
```

```
{
  animal_name: "Cheetah",
  nature: "harm",
  favorite_foods: ["meat", "gazelle", "rabbits"],
  care_taker_name: "Priya",
  life_span: 12,
  timestamp: new Date("2022-01-18"),
  expenses: 45000
},
{
  animal_name: "Giraffe",
  nature: "harmless",
  favorite_foods: ["leaves", "fruits", "grass"],
  care_taker_name: "Ravi",
  life_span: 25,
  timestamp: new Date("2023-05-30"),
  expenses: 35000
}
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69b13e10a03ea65b4b971700'),
    '1': ObjectId('69b13e10a03ea65b4b971701'),
    '2': ObjectId('69b13e10a03ea65b4b971702'),
    '3': ObjectId('69b13e10a03ea65b4b971703'),
    '4': ObjectId('69b13e10a03ea65b4b971704')
  }
}
```

5. Add 5 domestic-animal details to the collection named 'domestic_animals'. Each document consists of following fields as animal_name, gender (male or female), favorite_foods (meat, rabbits, deer etc) as array, animal_petname, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.

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```
> db.domestic_animals.insertMany([
  {
    animal_name: "Dog",
    gender: "male",
    favorite_foods: ["bones", "meat", "biscuits"],
    animal_petname: "Bruno",
    life_span: 13,
    timestamp: new Date("2021-06-15"),
    expenses: 5000
  },
  {
    animal_name: "Cat",
    gender: "female",
    favorite_foods: ["fish", "milk", "meat"],
    animal_petname: "Kitty",
    life_span: 15,
    timestamp: new Date("2020-09-10"),
    expenses: 3000
  },
  {
    animal_name: "Rabbit",
    gender: "male",
    favorite_foods: ["carrots", "grass", "lettuce"],
    animal_petname: "Bunny",
    life_span: 8,
    timestamp: new Date("2022-03-25"),
    expenses: 2000
  },
])
```

```
{
  animal_name: "Parrot",
  gender: "female",
  favorite_foods: ["seeds", "fruits", "nuts"],
  animal_petname: "Polly",
  life_span: 30,
  timestamp: new Date("2019-12-01"),
  expenses: 1500
},
{
  animal_name: "Cow",
  gender: "female",
  favorite_foods: ["grass", "hay", "grains"],
  animal_petname: "Daisy",
  life_span: 20,
  timestamp: new Date("2023-08-14"),
  expenses: 8000
}
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69b13e5aa03ea65b4b971705'),
    '1': ObjectId('69b13e5aa03ea65b4b971706'),
    '2': ObjectId('69b13e5aa03ea65b4b971707'),
    '3': ObjectId('69b13e5aa03ea65b4b971708'),
    '4': ObjectId('69b13e5aa03ea65b4b971709')
  }
}
```

6. Write a MongoDB query to display all documents available in wild_animals and domestic_animals.

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech

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Student name: J MADHAN KUMAAR

Date: 11-03-2026

School: SCOPE

Reg. no.: 23BCE8930

```
> db.wild_animals.find().pretty()
< {
  _id: ObjectId('69b13e10a03ea65b4b971700'),
  animal_name: 'Lion',
  nature: 'harm',
  favorite_foods: [
    'meat',
    'deer',
    'zebra'
  ],
  care_taker_name: 'John',
  life_span: 15,
  timestamp: 2021-03-10T00:00:00.000Z,
  expenses: 50000
}
{
  _id: ObjectId('69b13e10a03ea65b4b971701'),
  animal_name: 'Tiger',
  nature: 'harm',
  favorite_foods: [
    'meat',
    'rabbits',
    'deer'
  ],
  care_taker_name: 'Ravi',
  life_span: 18,
  timestamp: 2020-07-22T00:00:00.000Z,
  expenses: 60000
}
```

```
{
  _id: ObjectId('69b13e10a03ea65b4b971702'),
  animal_name: 'Elephant',
  nature: 'harmless',
  favorite_foods: [
    'sugarcane',
    'bananas',
    'grass'
  ],
  care_taker_name: 'John',
  life_span: 60,
  timestamp: 2019-11-05T00:00:00.000Z,
  expenses: 80000
}
{
  _id: ObjectId('69b13e10a03ea65b4b971703'),
  animal_name: 'Cheetah',
  nature: 'harm',
  favorite_foods: [
    'meat',
    'gazelle',
    'rabbits'
  ],
  care_taker_name: 'Priya',
  life_span: 12,
  timestamp: 2022-01-18T00:00:00.000Z,
  expenses: 45000
}
{
```

```
  _id: ObjectId('69b13e10a03ea65b4b971704'),
  animal_name: 'Giraffe',
  nature: 'harmless',
  favorite_foods: [
    'leaves',
    'fruits',
    'grass'
  ],
  care_taker_name: 'Ravi',
  life_span: 25,
  timestamp: 2023-05-30T00:00:00.000Z,
  expenses: 35000
}
> db.domestic_animals.find().pretty()
< {
  _id: ObjectId('69b13e5aa03ea65b4b971705'),
  animal_name: 'Dog',
  gender: 'male',
  favorite_foods: [
    'bones',
    'meat',
    'biscuits'
  ],
  animal_petname: 'Bruno',
  life_span: 13,
  timestamp: 2021-06-15T00:00:00.000Z,
  expenses: 5000
}
}
```

```
{
  _id: ObjectId('69b13e5aa03ea65b4b971706'),
  animal_name: 'Cat',
  gender: 'female',
  favorite_foods: [
    'fish',
    'milk',
    'meat'
  ],
  animal_petname: 'Kitty',
  life_span: 15,
  timestamp: 2020-09-10T00:00:00.000Z,
  expenses: 3000
}
{
  _id: ObjectId('69b13e5aa03ea65b4b971707'),
  animal_name: 'Rabbit',
  gender: 'male',
  favorite_foods: [
    'carrots',
    'grass',
    'lettuce'
  ],
  animal_petname: 'Bunny',
  life_span: 8,
  timestamp: 2022-03-25T00:00:00.000Z,
  expenses: 2000
}
}
```

Lab Sheet 6: MongoDB Basic commands

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```
{
  _id: ObjectId('69b13e5aa03ea65b4b971708'),
  animal_name: 'Parrot',
  gender: 'female',
  favorite_foods: [
    'seeds',
    'fruits',
    'nuts'
  ],
  animal_petname: 'Polly',
  life_span: 30,
  timestamp: 2019-12-01T00:00:00.000Z,
  expenses: 1500
}
{
  _id: ObjectId('69b13e5aa03ea65b4b971709'),
  animal_name: 'Cow',
  gender: 'female',
  favorite_foods: [
    'grass',
    'hay',
    'grains'
  ],
  animal_petname: 'Daisy',
  life_span: 20,
  timestamp: 2023-08-14T00:00:00.000Z,
  expenses: 8000
}
```

7. Write a MongoDB query to display only animal name and expenses in all the collection of the database

```
> db.wild_animals.find({}, { animal_name: 1, expenses: 1, _id: 0 })
< {
  animal_name: 'Lion',
  expenses: 50000
}
{
  animal_name: 'Tiger',
  expenses: 60000
}
{
  animal_name: 'Elephant',
  expenses: 80000
}
{
  animal_name: 'Cheetah',
  expenses: 45000
}
{
  animal_name: 'Giraffe',
  expenses: 35000
}
```

```
> db.domestic_animals.find({}, { animal_name: 1, expenses: 1, _id: 0 })
< {
  animal_name: 'Dog',
  expenses: 5000
}
{
  animal_name: 'Cat',
  expenses: 3000
}
{
  animal_name: 'Rabbit',
  expenses: 2000
}
{
  animal_name: 'Parrot',
  expenses: 1500
}
{
  animal_name: 'Cow',
  expenses: 8000
}
```

8. Write a MongoDB query to display domestic_animals whose life is a particular year

Lab Sheet 6: MongoDB Basic commands

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Faculty Name: Prof. S.Gopikrishnan

School: SCOPE

Student name: J MADHAN KUMAAR

Reg. no.: 23BCE8930

```
> db.domestic_animals.find({ life_span: 15 })
< {
  _id: ObjectId('69b13e5aa03ea65b4b971706'),
  animal_name: 'Cat',
  gender: 'female',
  favorite_foods: [
    'fish',
    'milk',
    'meat'
  ],
  animal_petname: 'Kitty',
  life_span: 15,
  timestamp: 2020-09-10T00:00:00.000Z,
  expenses: 3000
}
```

9. Write a MongoDB query to display wild_animals available under a particular care_taker

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech

Faculty Name: Prof. S.Gopikrishnan

Student name: J MADHAN KUMAAR

Date: 11-03-2026

School: SCOPE

Reg. no.: 23BCE8930

```
> db.wild_animals.find({ care_taker_name: "John" })
< {
  _id: ObjectId('69b13e10a03ea65b4b971700'),
  animal_name: 'Lion',
  nature: 'harm',
  favorite_foods: [
    'meat',
    'deer',
    'zebra'
  ],
  care_taker_name: 'John',
  life_span: 15,
  timestamp: 2021-03-10T00:00:00.000Z,
  expenses: 50000
}
{
  _id: ObjectId('69b13e10a03ea65b4b971702'),
  animal_name: 'Elephant',
  nature: 'harmless',
  favorite_foods: [
    'sugarcane',
    'bananas',
    'grass'
  ],
  care_taker_name: 'John',
  life_span: 60,
  timestamp: 2019-11-05T00:00:00.000Z,
  expenses: 80000
}
```

10. Write a MongoDB query to display animal name, favorite_foods and expenses details whose lifespan is more than 5 years.

Lab Sheet 6: MongoDB Basic commands

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Student name: J MADHAN KUMAAR

Date: 11-03-2026

School: SCOPE

Reg. no.: 23BCE8930

```
> db.wild_animals.find(
  { life_span: { $gt: 5 } },
  { animal_name: 1, favorite_foods: 1, expenses: 1, _id: 0 }
)
< {
  animal_name: 'Lion',
  favorite_foods: [
    'meat',
    'deer',
    'zebra'
  ],
  expenses: 50000
}
{
  animal_name: 'Tiger',
  favorite_foods: [
    'meat',
    'rabbits',
    'deer'
  ],
  expenses: 60000
}
{
```

```
{
  animal_name: 'Elephant',
  favorite_foods: [
    'sugarcane',
    'bananas',
    'grass'
  ],
  expenses: 80000
}
{
  animal_name: 'Cheetah',
  favorite_foods: [
    'meat',
    'gazelle',
    'rabbits'
  ],
  expenses: 45000
}
{
  animal_name: 'Giraffe',
  favorite_foods: [
    'leaves',
    'fruits',
    'grass'
  ],
  expenses: 35000
}
```

Lab Sheet 6: MongoDB Basic commands

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Student name: J MADHAN KUMAAR

Date: 11-03-2026

School: SCOPE

Reg. no.: 23BCE8930

```
> db.domestic_animals.find(
  { life_span: { $gt: 5 } },
  { animal_name: 1, favorite_foods: 1, expenses: 1, _id: 0 }
)
< {
  animal_name: 'Dog',
  favorite_foods: [
    'bones',
    'meat',
    'biscuits'
  ],
  expenses: 5000
}
{
  animal_name: 'Cat',
  favorite_foods: [
    'fish',
    'milk',
    'meat'
  ],
  expenses: 3000
}
{
  animal_name: 'Rabbit',
  favorite_foods: [
    'carrots',
    'grass',
    'lettuce'
```

```
    animal_name: 'Rabbit',
    favorite_foods: [
      'carrots',
      'grass',
      'lettuce'
    ],
    expenses: 2000
  }
  {
    animal_name: 'Parrot',
    favorite_foods: [
      'seeds',
      'fruits',
      'nuts'
    ],
    expenses: 1500
  }
  {
    animal_name: 'Cow',
    favorite_foods: [
      'grass',
      'hay',
      'grains'
    ],
    expenses: 8000
  }
}
animal>
```